CARSON KOHLBRENNER

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EDUCATION

Ph.D. in Robotics - University of Colorado Boulder — Aug 2024 – May 2030
Department: Computer Science

M.sc. in Autonomous Systems - University of Colorado Boulder — Aug 2024 – May 2025
GPA: 4.0/4.0
Department: Ann & H.J. Smead Aerospace Engineering Sciences
Thesis Title: "Procedurally Generated Whole-Body Artificial Skin for Robot Applications"

B.sc. in Aerospace Engineering - University of Colorado Boulder — Aug 2020 – May 2024
GPA: 3.9/4.0
Minor: Computer Science. *Awards:* Summa Cum Laude

PUBLICATIONS

Note: 'C' = conference paper and 'W' = workshop paper.

[C, 2026] **C. Kohlbrenner**, A. Soukhovei, C. Escobedo, A. Roncone, "**Design, Mapping, and Contact Anticipation with 3D-printed Whole-Body Tactile and Proximity Sensors**" in *IEEE International Conference on Robotics and Automation (ICRA).*

[C, 2026] H. Chen, **C. Kohlbrenner**, J. Kubik, L. Rustler, A. Dickhans, K. Bartunek, A. Roncone, H. Lee, M. Hoffmann, "**Toward Geometry-Scalable Whole-Body Touch for Humanoids: A 3D-Printed Conformal EIT Skin**" in *IEEE-RAS International Conference on Humanoid Robots (Humanoids).* [In Review]

[C, 2025] **C. Kohlbrenner**, C. Escobedo, S. Bae, A. Dickhans, A. Roncone, "**GenTact Toolbox: A Computational Design Pipeline to Procedurally Generate Context-Driven 3D Printed Whole-Body Artificial Skins**" in *IEEE International Conference on Robotics and Automation (ICRA).*

[W, 2026] **C. Kohlbrenner**, N. Pudasaini, W. Xie, N. Sivagnanadasan, N. Correll, A. Roncone, "**Egocentric Tactile and Proximity Sensors as Observation Priors for Humanoid Collision Avoidance**" in the *8th RoboTac Workshop at the IEEE International Conference on Robotics and Automation (ICRA).*

[W, 2026] **C. Kohlbrenner**, C. Escobedo, S. Ray, A. Dickhans, A, Soukhovei, N. Jackoski, L. Antieau, A. Roncone, "**Improving Sensing Coverage and Compliance of 3D-Printed Artificial Skins Through Multi-Modal Sensing and Soft Materials**" in the *Towards Large-Area Tactile Sensing Skins workshop at the International Conference on Robotics and Automation (ICRA).*

[W, 2026] H. Chen, **C. Kohlbrenner**, J. Kubik, L. Rustler, A. Dickhans, A. Roncone, H. Lee, M. Hoffmann, "**A 3D-Printed Electrical Impedance Tomography Tactile Sensor for Scalable Coverage**" in the *Towards Large-Area Tactile Sensing Skins workshop at the International Conference on Robotics and Automation (ICRA).*

[W, 2025] **C. Kohlbrenner**, M. Murray, Y. Zhang, C. Escobedo, T. Dunnington, N. Stevenson, N. Correll, A. Roncone, "**A Machine Learning Approach to Contact Localization in Variable Density Three-Dimensional Tactile Artificial Skin**" in the *Workshop on Touch Processing at the Conference on Neural Information Processing Systems (NeurIPS).*

[W, 2025] A. Soukhovei, **C. Kohlbrenner**, C. Escobedo, A. Roncone, "**Form-Fitting, Large-Area Sensor Mounting for Obstacle Detection**" in the *Humanoids Workshop on Advances in Contact-Rich Robotics (ConRich) at IEEE International Conference on Robotics and Automation (ICRA).*

ACADEMIC SERVICE

Workshops

- Rocky Mountain AI Interest Group (RMAIIG 2025)

Reviewer

- IEEE Robotics and Automation Letters [RA-L] (2025, 2026)
- IEEE-RAS 24th International Conference on Humanoid Robots [Humanoids] (2025)
- IEEE International Conference on Robotics and Automation [ICRA] (2024, 2025, 2026)

WORK EXPERIENCE

Teaching Assistant – University of Colorado Boulder, Boulder, CO — Jan 2024 – Present

Description: Facilitated undergraduate engineering labs for the following courses: ASEN 3801 - Orbital Dynamics and Controls, ASEN 4018/4028 - Aerospace Senior Projects, and CSCI 1300 - Introduction to Computer Science.

Senior ML / Robotics Engineer – Foxglove, San Francisco, CA — July 2025 – Jan 2026

Description: Created robotics demos that showcased Foxglove's data visualization software. Wrote tutorial articles and held workshops to teach customers how to use the Foxglove SDK for rapid data visualization.

Research Assistant – Human Interaction & Robotics Group, Boulder, CO — May 2024 – Sep 2024

Description: First master's student to be fully funded in the lab during a summer semester. Studied the integration of multi-modal sensing with an emphasis on the sense of touch. Led a team of three to create a new class of whole-body tactile sensors.

Assistant Engineer/Technician - Technology Applications Inc., Boulder, CO — July 2021 – Jan 2024

Description: Aided in the design and manufacturing of high-temperature thermal transfer straps for NASA and Army-funded SBIR programs. Performed and managed two full-company audits. Heavily involved in proposal writing.

AWARDS

Outstanding Paper Award, University of Colorado Boulder — Mar 2026
Student Travel Grant, University of Colorado Boulder — Mar 2025
Sewall Esteemed Scholars Recipient, University of Colorado Boulder — Aug 2020